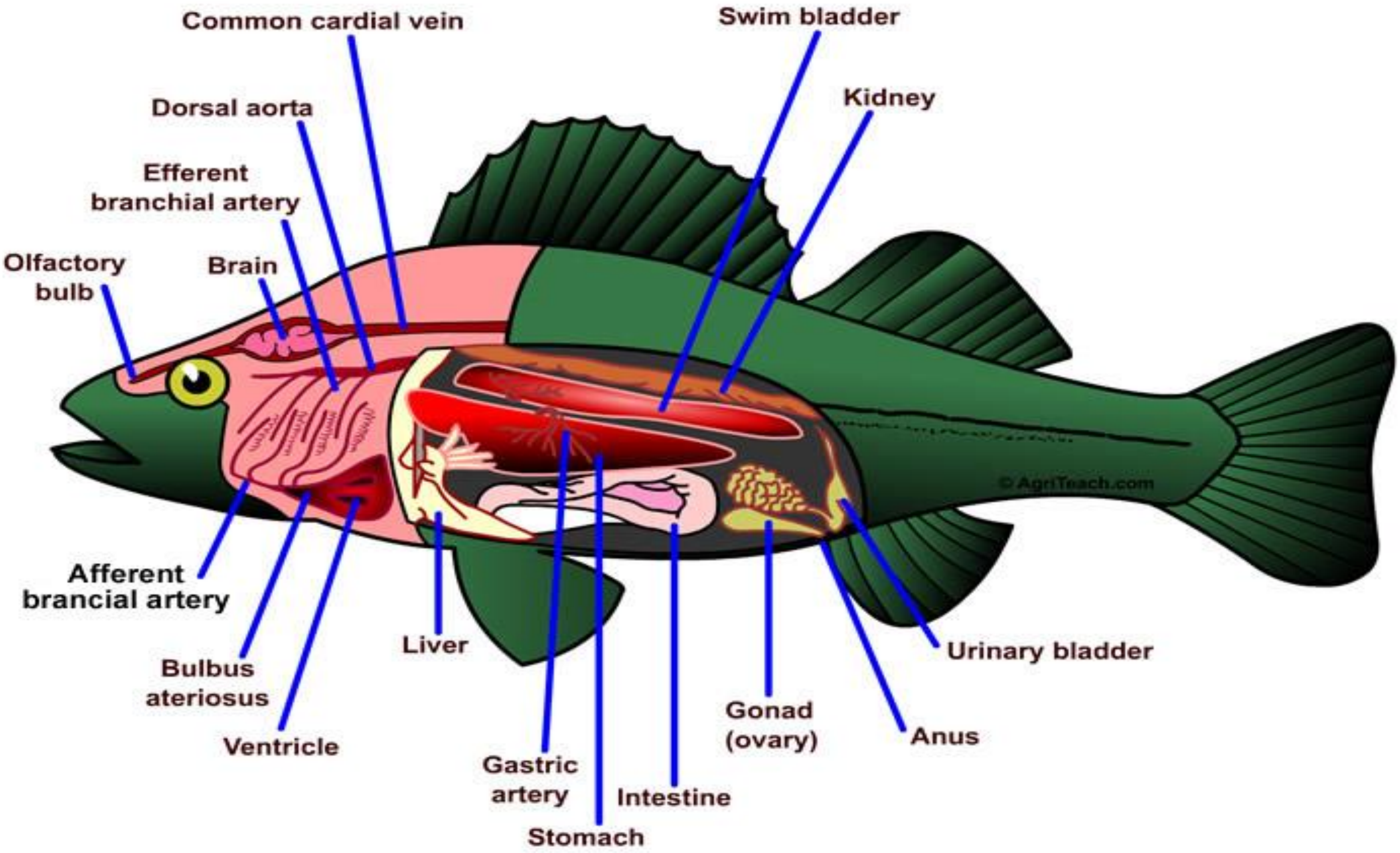
Fish Histology

Fish histology

Tissues of fishes are sometimes different from those of humans and other animals as they are aquatic creatures.

Basic Fish Anatomy



Digestive system

*** The digestive system of fish is basically composed of the alimentary canal and digestive glands (gastric and intestinal glands, liver and pancreas).**

1- Alimentary canal:

* It extends from the mouth to the anus

* The histological structure of the wall of the alimentary canal of fish is generally similar to that of higher vertebrates, consisting of ..

a) Mucosa
- mucosal epithelium
- basal lamina
- lamina propria
- muscularis mucosa

b) Submucosa → **stratum compactum**
stratum granulosum

c) Muscularis → **circular muscle layer**
longitudinal muscle layer

d) Serosa → **tella subserosa**
serous membrane

- **The lips:**

*** They are composed of the epidermis and the dermis.**

***The epidermis consists of stratified squamous epithelium in which club cells, mucous cells, taste buds and wandering cells are found.**

*** The dermis consists of reticular fibers, collagen fibers and elastic fibers. It contains blood vessels and nerve fibers.**

*** The lips are not true in fish as in mammals as they do not contain any muscle tissue.**

- **Buccal cavity:**

The mucosa is composed of stratified squamous epithelium and the lamina propria forms the core of the finger like subepithelial papillae which project into the epithelium.

The submucosa is composed of loose C.T.

The muscularis is composed of bands of longitudinally disposed striated muscle fibers.

The mucosa of the roof of the buccal cavity contains fewer taste buds as compared with its floor.

Fish in contrast to mammals do not chew their food and they therefore do not need cheeks

• Tongue:

* The tongue is generally poorly-developed in fish.

* It usually does not contain muscle tissue, and so it depends on the passive motion.

·Pharynx:

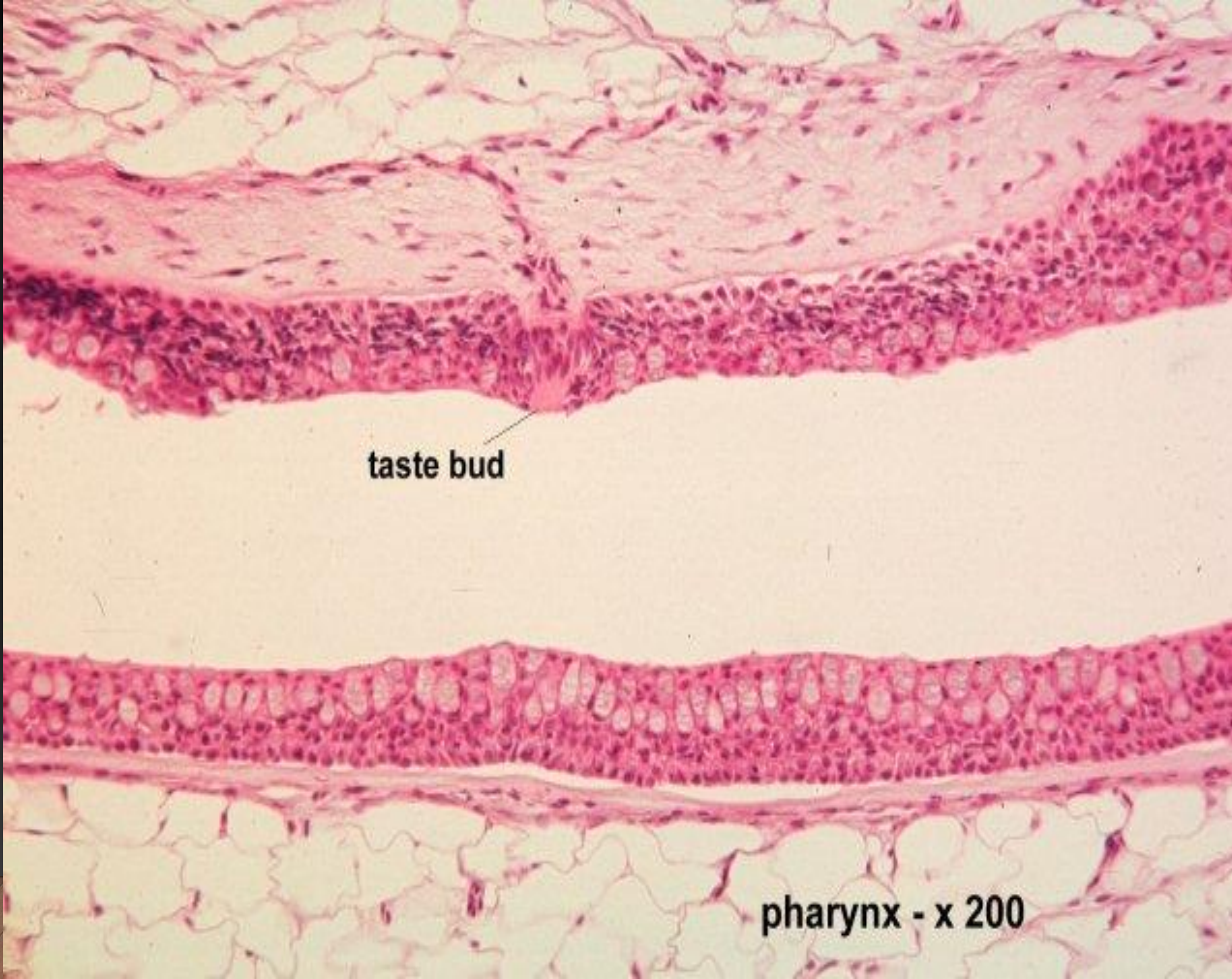
* The mucosa shows shallow folds of stratified epithelium.

* Epithelial cells of the outer layer are flat, but those at the base are columnar.

* Mucous cells are present among these cells being especially numerous at the bottom part of the folds.

* Lamina propria is found but muscularis mucosa is absent.

* The muscularis is composed of two layers of striated muscle, a thick outer layer of circular muscle and a thin inner layer of longitudinal muscle.


taste bud
pharynx - x 200

- **Esophagus:**

*** Its epithelium is stratified at the anterior portion but becomes single-layered towards the transition to the stomach.**

*** Mucous cells are numerous in the mucosal epithelium of the posterior end of the esophagus.**

*** Muscularis mucosa is generally absent.**

***The submucosa is composed of C.T. somewhat denser than that of the lamina propria.**

***The muscularis consists of striated muscle in the anterior portions and gradually changes to smooth muscle near the transition to the stomach.**

*** Tunica serosa is usually two cells thick, it corresponds to the peritoneal epithelium of the walls of the abdominal cavity.**


Esophagus - x 200


Transition esophagus-stomach - x 400
Serous cardiac gland

- **Stomach:**

* 1 to 3 segments can be distinguished in the stomach of different species of fish.

* The stomach is composed of the cardiac portion, blind-sac (corpus) and pyloric portion.

* The mucosal epithelium of each portion is single-layered and folded.

* The folds at the cardiac portion are shallow, but those at the corpus and pyloric portion become deeper.

* Epithelial cells of the cardiac portion are cuboidal in shape, and those of the rest of the stomach epithelium are high columnar. The nucleus is generally located in the basal region of the cell.

* The secretory cells that compose the gastric glands are polygonal in shape and contain granules. They occupy the bottom of the gastric gland while mucous cells are found in the neck region.

* It is more often that, muscularis mucosa is absent.

* Muscularis is composed of smooth muscles arranged in a thick inner coat of circular muscle and a thin outer coat of longitudinal muscles which may be absent in some species.


Serous cardiac gland - x 400

Pyloric Caeca - H&E x 160

- **Intestine:**

* The small intestine in fish is difficult to distinguish (duodenum, jejunum and ileum).

* The small intestine is lined by a simple columnar epithelium characterized by central nuclei and scattered goblet cells.

* The muscularis mucosa is usually missing.

* Brunner's gland have not been reported in fish.

* The musculature throughout the intestine is usually smooth muscle.

*** The serosa consists of a thin layer of C.T., covered with a simple epithelium continuous with the mesentery.**

*** Also it is difficult to distinguish the large intestine in fish.**

*** <u>Usually the large intestine of fish is not differentiated into colon and rectum</u>.**

*** Most fish have a separated anus rather than a true cloaca. The anus of fish has a muscular sphincter. Its epithelium changes from columnar to stratified squamous.**

*** A true cloaca may be found in some species.**

2- Liver:

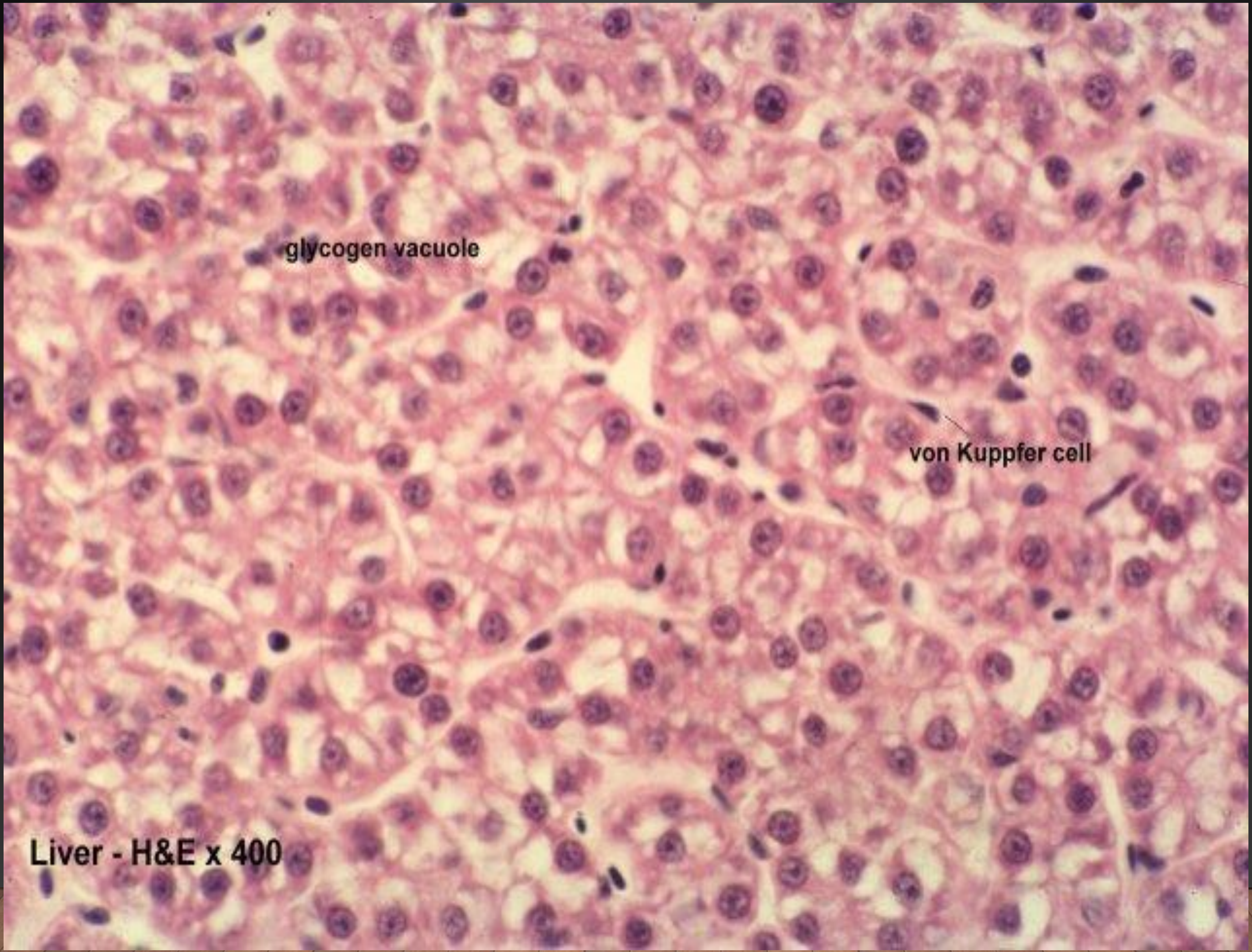
* The hepatic cells do not display definite pattern of arrangement similar to that of fowl.

* These cells are mostly polygonal, however hepatic cells surrounding blood vessels and sinusoids are columnar.

* Each hepatic cell contains fine, granulated basophilic cytoplasm with large centrally located nucleus.

* Columnar hepatic cells have basal nuclei.

An admixture of hepatocellular and pancreatic tissues is present in many fish forming a hepatopancreas.


glycogen vacuole
von Kuppfer cell
Liver - H&E x 400


Arteriole
Bile Duct

3- Pancreas:

*** The pancreas is present in all fish.**

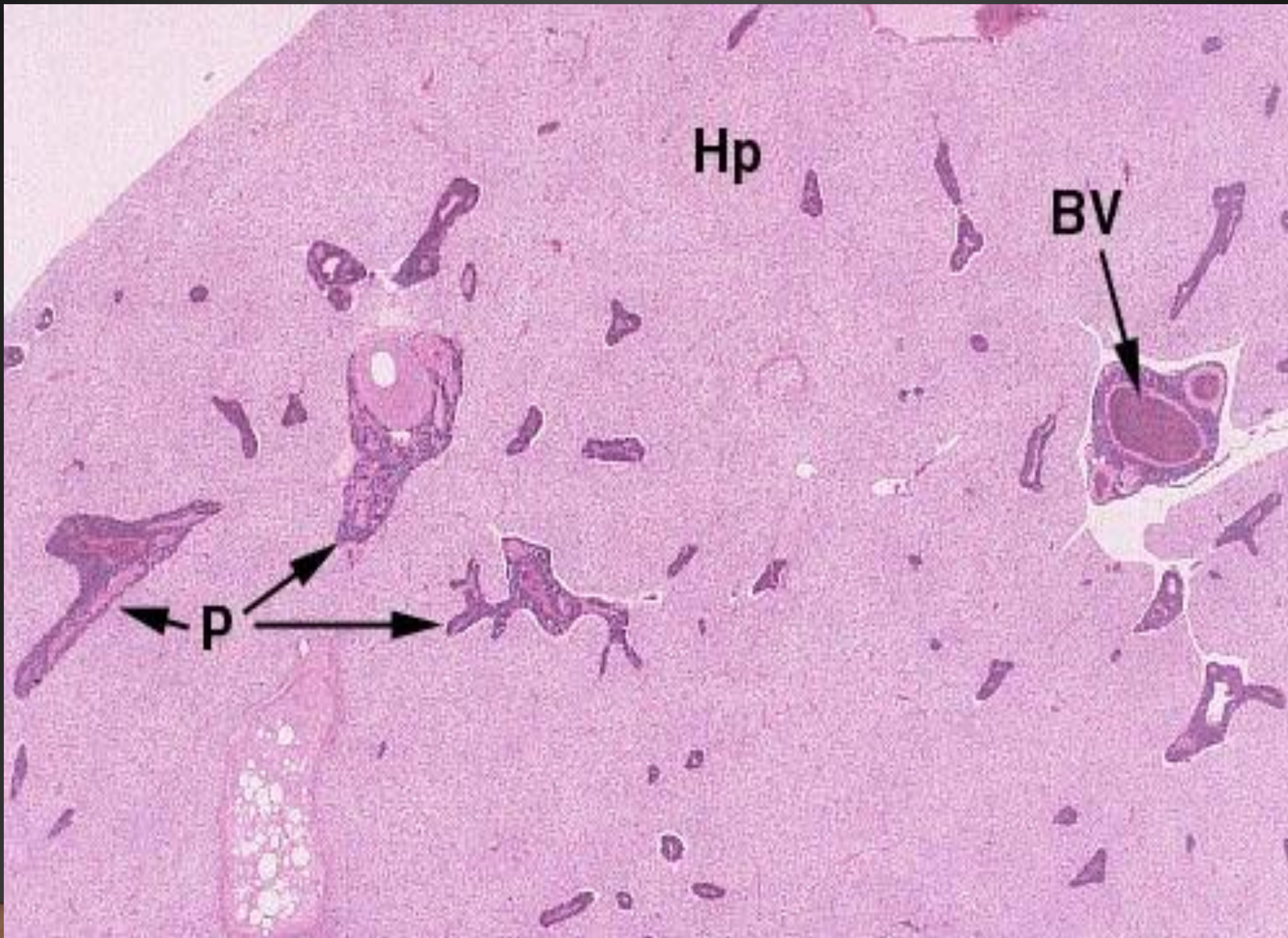
*** It may be massive (compact) or diffuse.**

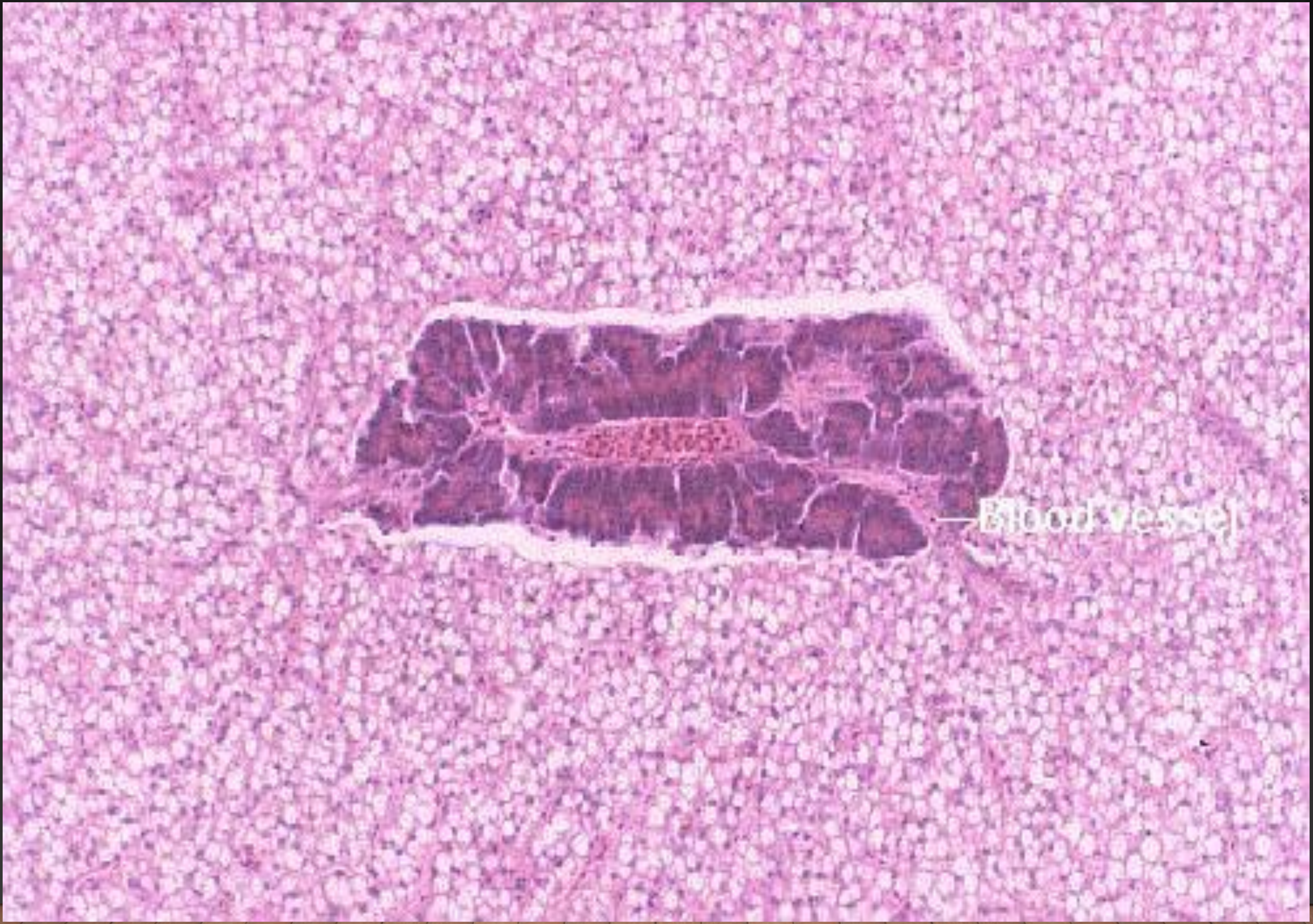
*** Many fish possess a pancreas that has an intimate association and admixture of cells, with the spleen or liver. This combined organ is termed "*splenopancreas* or *hepatopancreas*", respectively.**

*** The endocrine pancreas is not as diffusely distributed as the exocrine pancreas, it is located near the liver and the spleen.**


Acinar cells
Pancreas -
H&E x 400


Acinar cells


Hp
BV
P


Blood vessel

Respiratory system

- **The respiratory system of fish is composed of gills.**
- **The epithelium is thin to allow gas exchange.**

The gills are also responsible for regulating the exchange of salt and water and play a major role in the excretion of nitrogenous waste products

Structure of the gills:

- **Primary lamellae:**

*** The primary lamella consists of cartilaginous support, vascular system, and multilayered epithelium.**

*** Squamous cells with fingerprint-like microridges exist as the outermost layer of the primary lamellar epithelium.**

*** The primary lamellar epithelium contains many mucous cells and chloride cells.**

Secondary lamellae:

* Numerous semicircular secondary lamellae are lined up along both sides of the primary lamella.

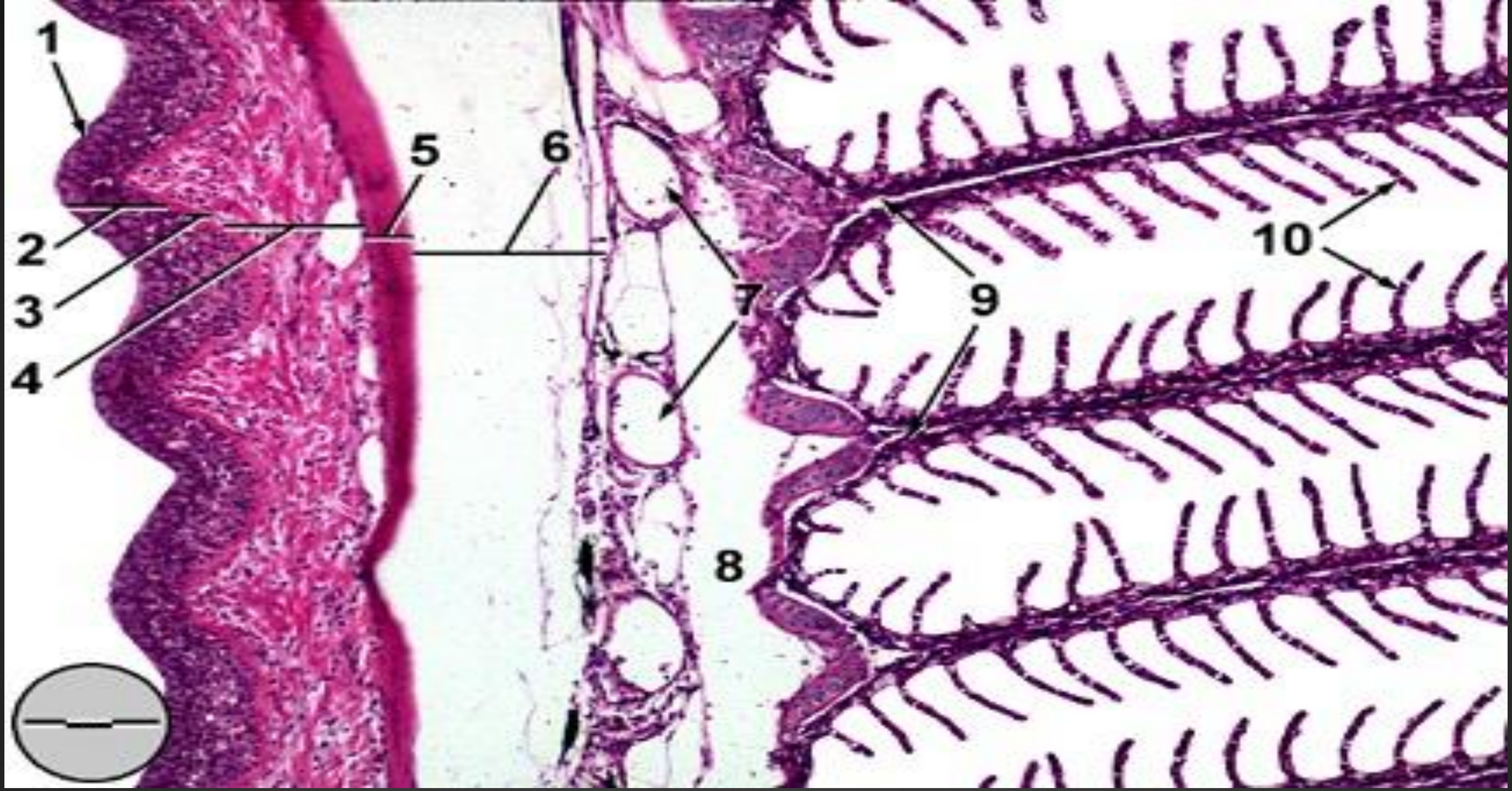
* the secondary lamellar epithelium consists of double layer of epithelial cells

* The outer layer is termed *respiratory epithelial cells* which have sparse and small microvilli and marginal microridges on the apical surface.

* The inner layer supports epithelial cells sitting on a basement membrane.

* The secondary lamellar epithelium is supported by many pillar cells.

*** Chloride cells** are seen on the basement of the secondary gill lamella. These cells are acidophilic and exist **generally in marine** fish and rarely in freshwater fish.



Gill arch: sagittal section :(Bouins, H&E). 1. gill raker 2.mucosal epithelium 3. basement membrane 4. submucosa 5. bone 6.adipose tissue 7. efferent branchial arterioles 8. afferent branchial artery 9.primary lamellae 10. secondary lamellae .

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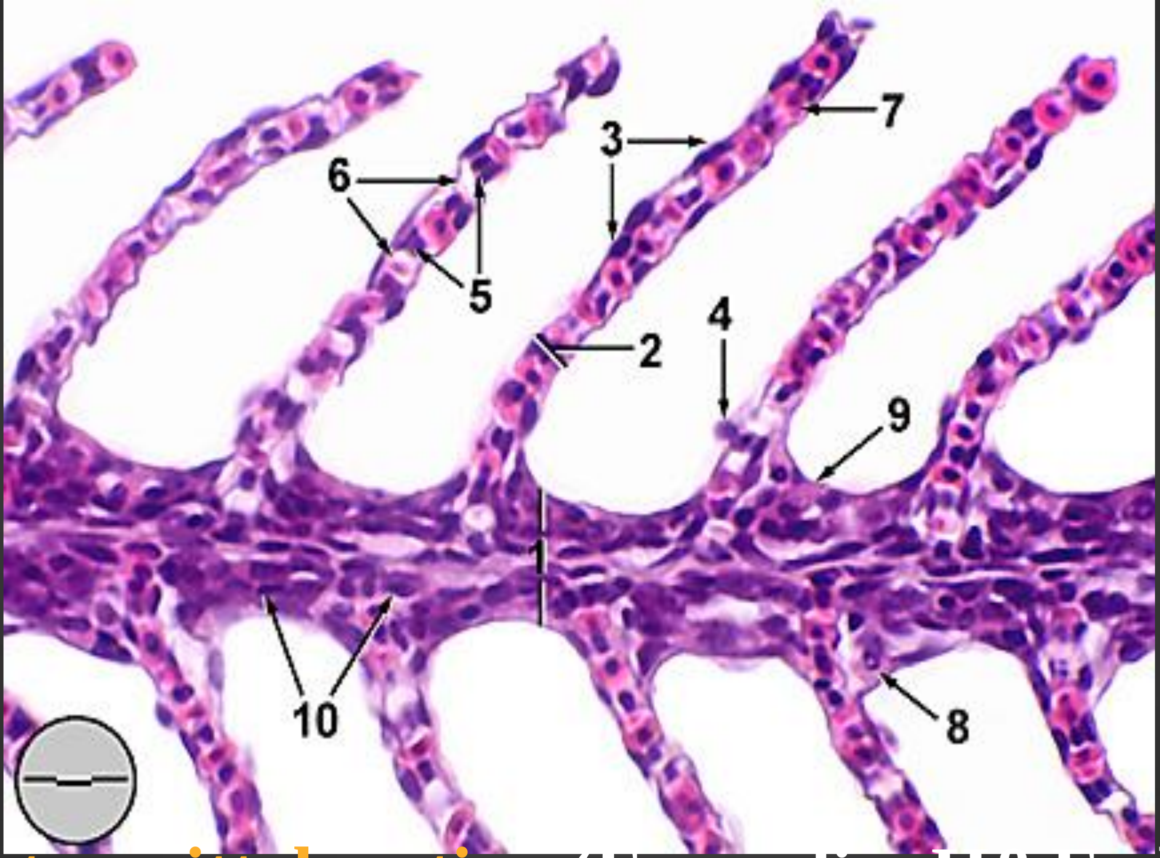


Gill filament, sagittal section (Formalin, H&E,).

1.primary lamella 2. secondary lamella

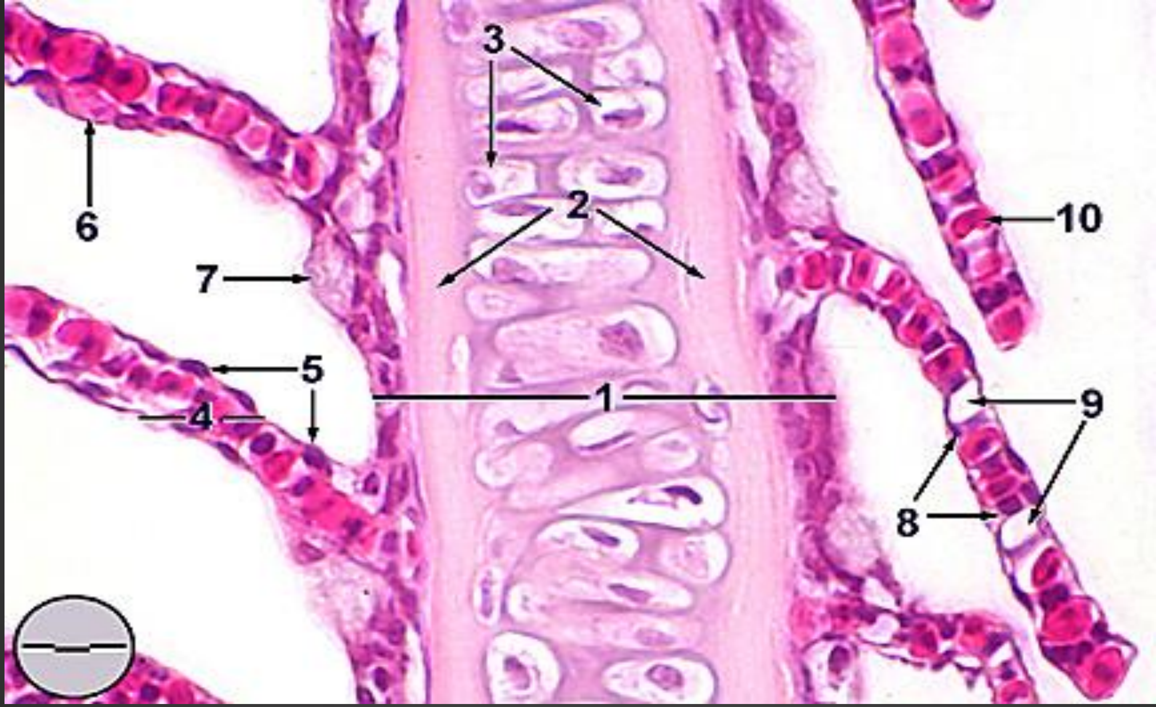
3.epithelial cell 4. mucous cell

5.pillar cell 6. lacuna (capillary lumen)

7.erythrocyte within capillary lumen

8.chloride cell 9. rodlet cell

10.undifferentiated basal cell.



Gill filament, sagittal section through cartilaginous support

(Formalin, H&E,). 1. primary lamella 2. extracellular cartilaginous matrix 3. chondrocytes 4. secondary lamella 5. epithelialcell 6. mucous cell 7. chloride cell 8. pillar cell 9. lacuna (capillary lumen)

10. red blood cells within lacuna

Urinary system

* Osmoregulation of the body fluid is a very important physiological function in aquatic animals.

* Kidney of fresh water fish excrete extra body water and that of marine fish excretes salts.

* The external form of fish kidney varies according to species.

* Fish nephron is devoid of the thin segment of Henle found in the nephron of higher vertebrates.

The nephron is followed by the collecting duct and ureter

* In some fish species a sac-like enlarged portion is detected in the posterior region of the ureter. This portion is called the urinary bladder, but it is not similar to that of higher vertebrates.

* The renal corpuscle is composed of a glomerulus and its capsule which has an outer fibrous layer and an inner flattened epithelium.

* The capsular epithelium is continuous with the renal tubular epithelium.

In the capsular epithelium, simple squamous cells with solitary cilia are delimited by marginal microvilli

* Mesangial cells fill the space between the capillary loop of the glomerulus.

* The glomerular capillary wall has 3 components: podocytes, basal lamina, and endothelial cells.

* The basal lamina beneath the podocytes is thicker in seawater fish than in freshwater fish.

* Extraglomerular cells are found in the walls of the afferent arterioles. These cells contain secretory granules which secrete renin hormone.

* Renal tubules are thin and short in the neck segment and consists of a single layer of low epithelial cells with long cilia.

* The proximal convoluted segment is divided into 2 parts, segment I and II.

* In segment I the tubular epithelium is composed of cuboidal cells with cilia and densely arranged microvilli on the apical surface.

* In segment II the cells are more taller (columnar), cilia and microvilli are found in the tubular lumen, with infolding of the basal membrane of the cells. Also the mitochondria are abundant in these cells.

* The intermediate segment is a specialized portion of segment II. It is well developed in carp kidney but absent in several fish species.

*** The epithelium become lower and more cuboidal in the intermediate segment, and the brush border becomes intermittent and ciliated.**

*** The distal convoluted tubules occur only in kidneys of freshwater fishes. Its epithelium is less eosinophilic and have no brush border.**

- **Ureter :**

*** It is lined by columnar epithelium.**

*** The middle layer is a C.T (lamina propria) and circular fibers of smooth muscle.**

*** The outer layer is C.T (tunica adventitia).**


tubules
Glomerulus
hematopoietic
tissue
Kidney - H&E x 400


Kidney - H&E x 400
tubules

Reproductive system

1- The ovary:

*** The mature ovaries represent as 70% of the total body weight.**

*** They are suspended from the abdominal wall by a mesentery and appear as a small cluster of minute orange-white spheres in the immature fish.**

*** Oogonia, the cells which are beginning to mature, are surrounded by a single layer of small epithelial cells. This aggregate of ova and epithelial cells is known as the ovarian follicle.**

*** The epithelial cells grow as the ovum grows and are separated from it by a gradually thickening hyaline capsule called Zona pellucida.**

* The granulosa cells are responsible for nourishing the ovum and secreting its yolk.

* In many species several generations of ova may be found in different stages of development.

* The ripe oocytes undergo hydration resulting in transparency and increase in volume. This leads to the rupture of the follicle and the eggs lie in the lumen of the ovary before passing out through the oviduct.

* The spent ovary is characterized by an extremely thickened theca and empty follicles. Large numbers of germ cells and resting oocytes are present forming the reserve for the next season.

* Soon after spawning has finished, resorption of the unshed mature oocytes is initiated by invasion of macrophages and melanomacrophages

(2) The testis:

* They are paired organs, suspended by mesentries from the dorsal abdominal wall

* They vary in size, they may approach 12% of the total body weight.

* There is a main collecting duct for genital secretions from the testis (the vas deferens) which conducts mature spermatozoa to an excretory meatus at the urinary papilla.

* The testis are a series of tubules or blind sacs, (the semineferous tubules) which are lined by spermatogenic epithelium.

* Spermatozoa attach to the surface of the pyriform nourishing cells of the seminiferous epithelium known as Sertoli cells until be ready to release

* Interstitial cells of the testis found in the fibrous supporting tissue or in the basement membrane of the seminiferous tubules. These cells secrete testosterone hormone.


Testes

Integumentary system

* The skin of fish lacks a horny layer.

* It also lacks multicellular glands such as sweat glands and sebaceous glands.

* It has mucous cells which are unicellular glands.

* The skin is the primary barrier against the environment, its layers are : 1- cuticle 2- epidermis

3- basement membrane 4- dermis 5-hypodermis

* The epidermis and the dermis contain some accessory organs, such as sensory receptors, scales, mucous glands, venom glands and luminous organs, which are specific to fish species.

1- Cuticle:

* It is 1 μm thick.

* It is mucopolysaccharide layer.

* It is normally secreted by epithelial surface cells rather than by goblet mucous cells.

* This layer contains specific immunoglobulins and lysozyme and free fatty acids, all of which are believed to have antipathogen activity.

2- <u>Epidermis:</u>

*** It is a stratified squamous epithelium covering the body surface and investing the tail and fins.**

*** It is a living layer able to divide mitotically at the level of all cell layers (even squamous layer).**

*** Many types of cells are found in the epidermis in addition to the epithelial cells as:**

- **<u>Mucous cells:</u>**

*** They are differentiated from basal cells, and gradually approach the surface region, increasing its size and after holocrine secretion at the surface, they fall off the epidermis.**

- **club cells:**

*** They are large ovoid or slender cells with eosinophilic cytoplasm.**

*** The cell has one or several nuclei in the center, while only a small amount of usual cytoplasm can be seen just around the nucleus.**

*** These cells are found in the lower and middle layers of the epidermis.**

*** They are secretory cells, and in some species secrete a potent alarm substance.**

- **Malpighian cells:**

*** They are rounded cells, very similar in structure at all levels except at the outermost, where they are horizontal and flattened.**

- **Granule cells:**

- **Migratory cells:**

*** Free cells as lymphocytes and macrophages.**

*** In mammals, there is a skin immune system independent of the interior one, but this does not have been confirmed in fish.**

3- Basement membrane:

- Similar in structure to the basement membrane of mammalian skin.

4- Dermis:

*** It is composed of two layers.**

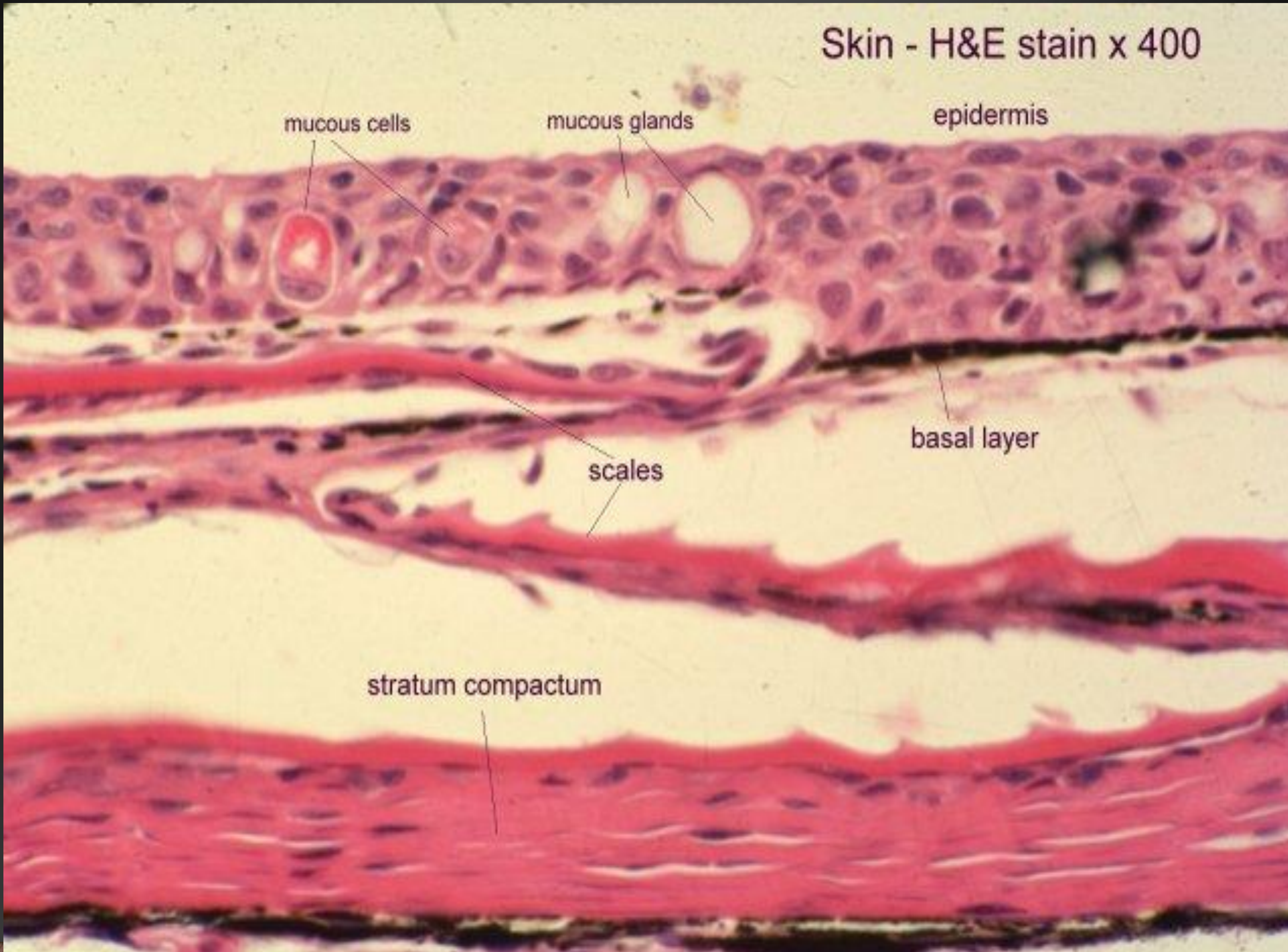
*** The upper *stratum spongiosum* which is a loose network of collagen and reticular fibers, continuous with the epidermal basement membrane and containing the pigment cells (chromatophores), mast cells and cells of the scale beds and also the scales.**

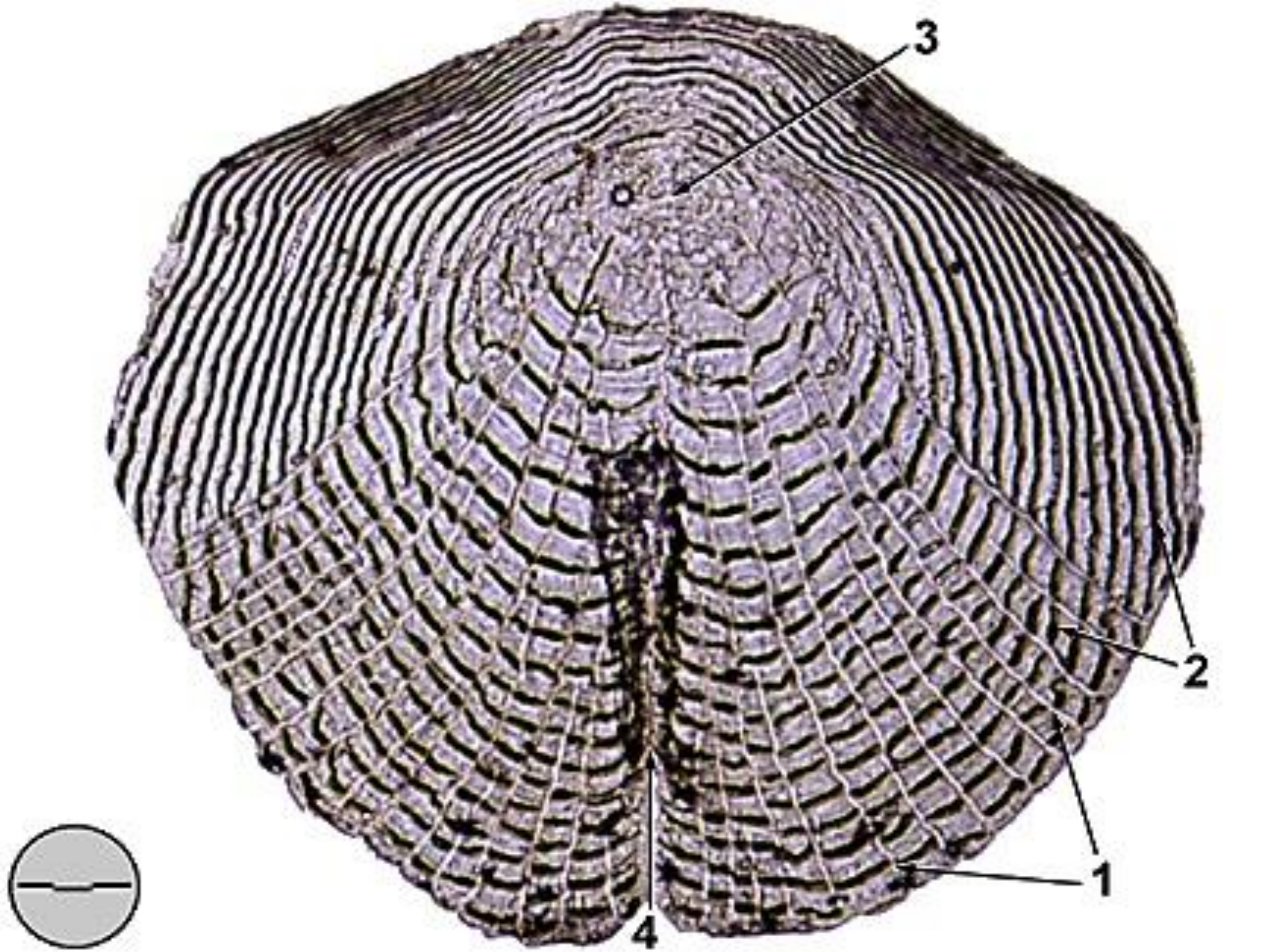
*** The second layer is *stratum compactum*, which is collagenous dense matrix providing the structural strength of the skin**

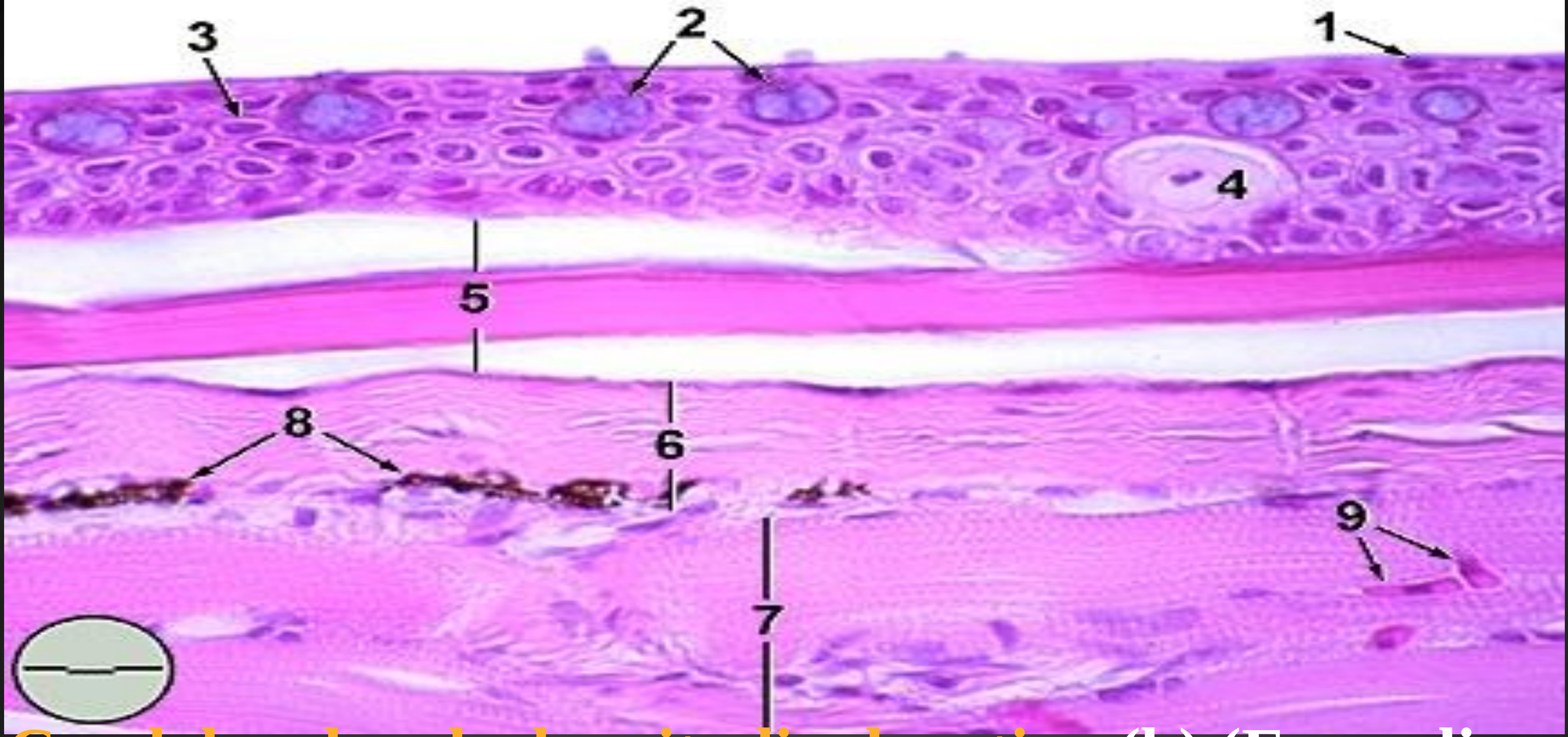
* **The scales consist of collagen fibers interspersed with a matrix of albuminoid materials with deposition of hydroxy apatite crystals.**

5- Hypodermis:

* **It is a looser, adipose, tissue which is more vascular than the dermis and a frequent site of development of infectious processes.**


Skin - H&E stain x 400
mucous cells
mucous glands
epidermis
basal layer
scales
stratum compactum


3
2
2
1
1
4



Caudal peduncle, longitudinal section (b) (Formalin, H&E) 1. squamous epithelial cells 2. mucous cells (expelling mucus) 3. cuboidal epithelial cells 4. alarm cell 5. scale pocket w/ scale 6. dermis (stratum compactum) 7. skeletal muscle

8. chromatophores (melanocytes) 9. red blood cells

THANK YOU